



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Yash Vilas Karande
Roll No. :	17
Experiment No.:	5
Title of the Experiment:	Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making
Date of Performance:	12/08/26
Date of Submission:	19/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi P.N.**

Signature :

Date :



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Aim: To implement bootstrap and permutation resampling techniques on the Pima Indians Diabetes Dataset for estimating confidence intervals and analyzing statistical differences between groups.

Objectives

After completing this experiment, the student will be able to:

1. Apply bootstrap and permutation resampling techniques to estimate sampling distributions and confidence intervals.
2. Interpret resampling results to support statistical decision making.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**. The Outcome variable indicates whether the patient is diabetic.

Procedure

1. Load the Pima Indians Diabetes Dataset.



2. Select a numerical variable such as Glucose or BMI.
3. Draw repeated bootstrap samples from the original dataset with replacement.
4. Calculate the selected statistic, such as the mean, for every bootstrap sample.
5. Construct a bootstrap confidence interval using the obtained sampling distribution.
6. Divide the data into two groups based on Outcome.
7. Calculate the observed difference between the group means.
8. Randomly permute the group labels repeatedly.
9. Calculate the difference in group means for every permutation.
10. Compare the observed difference with the permutation distribution and interpret the result.

Description of the Experiment

Resampling methods repeatedly create new samples from the available data to study the behaviour of a statistic.

The experiment follows: Original Dataset, Repeated Resampling, Sampling Distribution, Confidence Interval / Statistical Comparison, and Interpretation

Two resampling techniques are implemented:

- **Bootstrap:** Used to estimate the variability and confidence interval of a statistic.
- **Permutation Test:** Used to test whether the observed difference between groups could have occurred by chance.

Detailed Description of Statistical Methods

Bootstrap Resampling

Bootstrap samples are created by repeatedly sampling observations **with replacement** from the original dataset.

For each bootstrap sample, a statistic such as the mean is calculated.



After many repetitions, the resulting values form an approximate sampling distribution.

A confidence interval can then be obtained from this distribution.

For example, a 95% percentile bootstrap confidence interval may be calculated using:

$$[Q_{97.5}]$$

where $Q_{2.5}$ and $Q_{97.5}$ are the 2.5th and 97.5th percentiles of the bootstrap distribution.

Permutation Test

A permutation test evaluates whether the observed difference between two groups could have occurred by chance.

For example:

$$\text{Difference in Mean Glucose} = \bar{x}_{\text{diabetic}} - \bar{x}_{\text{non-diabetic}}$$

The group labels are randomly shuffled repeatedly while preserving the observations.

The resulting differences form the **null distribution**.

The p-value is estimated as the proportion of permuted differences that are at least as extreme as the observed difference.

Output:

```
Output ... Filter (e.g. text, lexcludeText, t... Code
[Running] python -u "e:\MYDATA\STATS\exp5.py"
===== DATASET INFORMATION =====
Dataset shape: (768, 9)

First 5 rows:
  Pregnancies  Glucose  BloodPressure  ...  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72  ...              0.627      50         1
1            1       85             66  ...              0.351      31         0
2            8      183             64  ...              0.672      32         1
3            1       89             66  ...              0.167      21         0
4            0      137             40  ...              2.288      33         1

[5 rows x 9 columns]

===== ORIGINAL DATA =====
Sample size: 768
Original mean Glucose: 120.895

===== BOOTSTRAP RESULTS =====
Number of bootstrap samples: 5000
Bootstrap mean: 120.898
Bootstrap standard error: 1.147
95% Bootstrap Confidence Interval: (np.float64(118.712), np.float64(123.125))

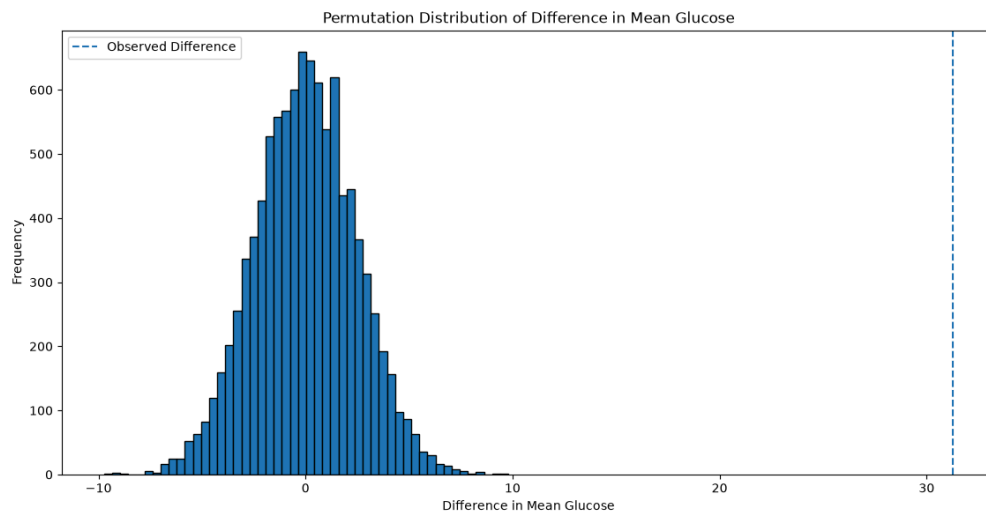
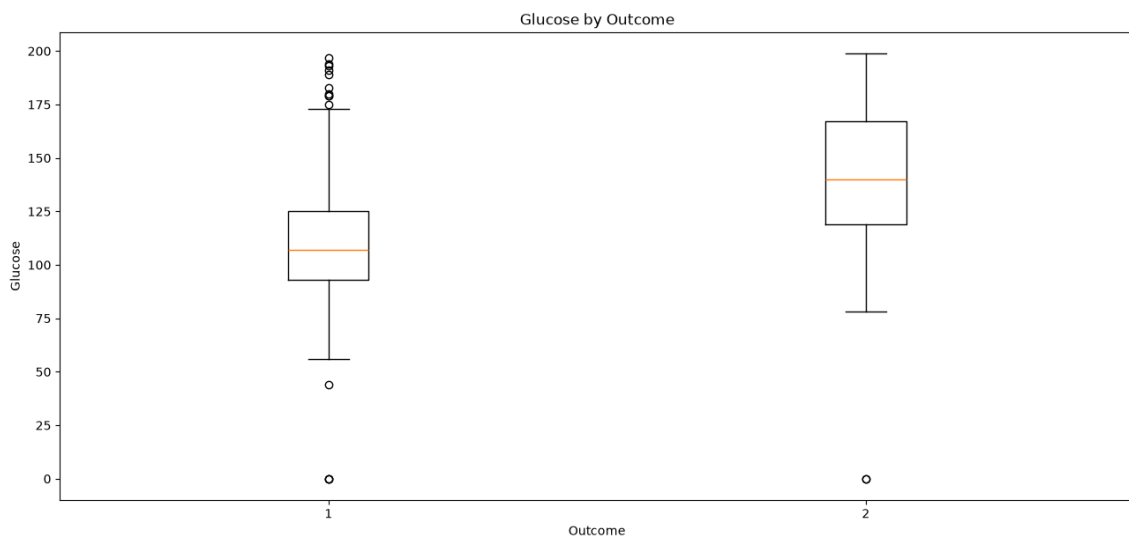
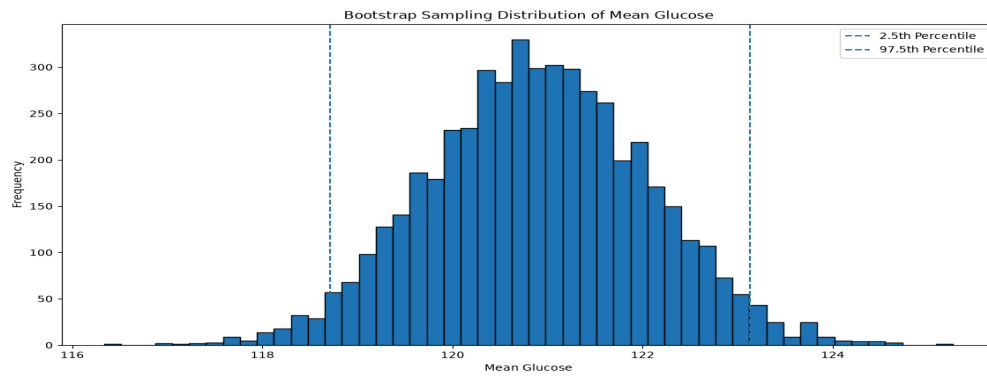
===== GROUP INFORMATION =====
Outcome 0 (No Diabetes): 500
Outcome 1 (Diabetes): 268

===== GROUP MEANS =====
Mean Glucose - Outcome 0: 109.98
Mean Glucose - Outcome 1: 141.257
Observed Difference (1 - 0): 31.277
```



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Conclusion

Bootstrap and permutation resampling techniques were implemented on the Pima Indians Diabetes Dataset. Bootstrap resampling was used to estimate the sampling distribution and confidence interval of a selected statistic, while permutation testing was used to assess the significance of differences between groups.

Questions to be Answered by Students

1. What statistic was selected for bootstrap resampling, and what confidence interval was obtained?

ANS: The mean was selected for bootstrap resampling. A 95% confidence interval was obtained for the population mean.

2. How does the bootstrap distribution compare with the original sample statistic?

ANS: Bootstrap distribution: The bootstrap distribution was centered close to the original sample mean, showing that the estimate was reasonably stable.

3. What was the observed difference between the selected groups?

ANS: Observed difference: The observed difference was the difference between the mean values of the two selected groups (Outcome 0 and Outcome 1)

4. Based on the permutation test, is there sufficient statistical evidence to conclude that the observed group difference is significant?

ANS: Permutation test: Based on the p-value, there was sufficient statistical evidence to conclude that the observed group difference was statistically significant if the p-value was less than 0.05.

5. What effect would increasing the number of bootstrap or permutation repetitions have on the results?

ANS: Increasing repetitions: Increasing the number of bootstrap/permutation repetitions makes the estimated distribution and p-value/confidence interval more stable and accurate, with less random variation.



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